Deep Double Descent

When bigger models, longer training, and more data can hurt

COMP34312 Mathematical Topics in Machine Learning | Ayush Kumar, Jacob Yip, Harshita Dassani

The University of Manchester

Objective: explain why test error can peak near interpolation and then fall again

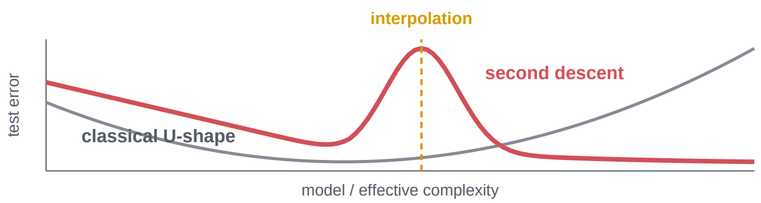
Problem: classical U-shaped overfitting intuition is incomplete for modern interpolating models

Approach: compare test and train error while changing width, epochs, samples, noise, optimizer, and regularization

Contribution: Nakkiran et al. unify model-wise, epoch-wise, and sample-wise effects using effective model complexity

1. The classical story is not enough

Lecture 5.3 introduces the surprise. The paper asks what controls the bad regime, and why the same pattern appears when we vary width, training time, or sample count.



2. Mathematical anchor: effective model complexity

critical regime: EMC_D,eps(T) is approximately n

R(f) = E_(x,y)~D [loss(y, f(x))]

Rhat_S(f) = (1/n) sum_i loss(y_i, f(x_i))

EMC_D,eps(T) = max { m : E_{S~D^m}[train_error(T(S), S)] <= eps }

- T is the full training procedure: architecture, optimizer, epochs, data augmentation, labels, and regularization.
- The threshold is not simply parameter count equals sample count.
- Changing epochs or data size can move the same system into or out of the peak.

3. Interpretation: a local danger zone

EMC << n
underfit
more fitting ability usually helps

EMC ~= n
critical
barely fits; noise or misspecification can dominate

EMC >> n
overparameterized
many interpolants; optimizer bias can matter

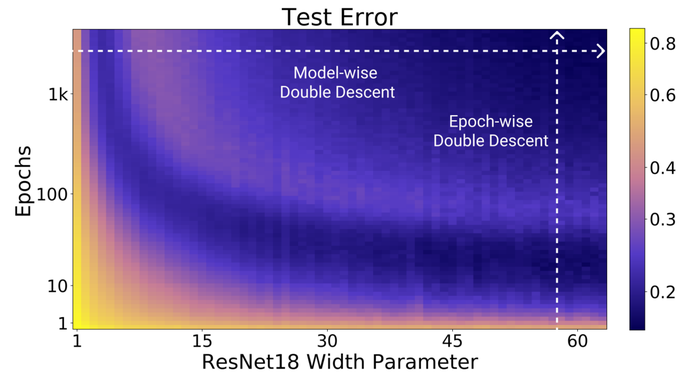
Proposed intuition, not a theorem: near the threshold, noise is costly; far beyond it, many interpolants make optimizer bias matter.

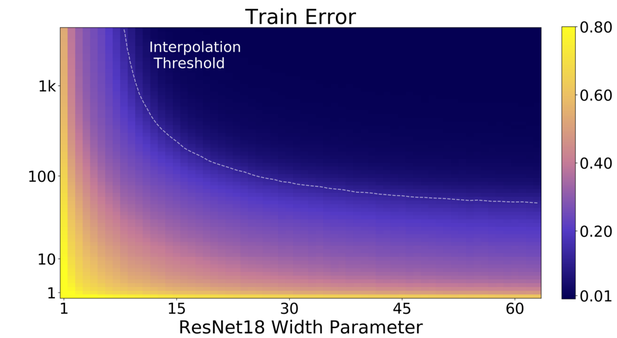
Double descent refines overfitting; it does not say overfitting is good.

4. Main empirical evidence from Nakkiran et al.

ResNet18 on CIFAR-10 with 15% label noise

Width and training time move the system through the same interpolation frontier.





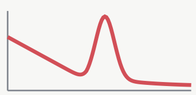
Train error companion: dashed curve marks interpolation threshold.

Test error: the bright ridge is the bad generalization zone.

Adapted from Nakkiran et al. arXiv:1912.02292. Horizontal slices show model-wise double descent; vertical slices show epoch-wise double descent.

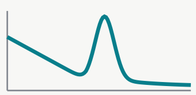
5. Three routes through the same critical regime

Model-wise



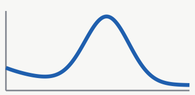
horizontal slice of heatmap

Epoch-wise



vertical slice of heatmap

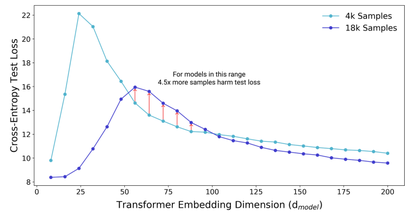
Sample-wise



fixed model; n can move into peak

Unifying claim: **the peak appears when EMC_D,eps(T) and n are close.**

6. More data can locally hurt



IWSLT Transformer: 18k samples can be worse than 4k for some widths.

This is local non-monotonicity, not the claim that data is bad overall.

7. Critical view and practical message

- Bias-variance is not false; the monotone overfitting story is incomplete.
- Bigger, longer-trained, or more-data systems can still be worse inside the critical interval.
- EMC is an empirical organizing lens, not a complete deep-net theory.

Takeaway

Generalization depends on the trained system, not model size alone. Use validation, regularization, early stopping, or enough scale to avoid the peak.

References

Nakkiran et al. Deep Double Descent, arXiv:1912.02292. Belkin et al. PNAS 2019. Zhang et al. ICLR 2017.

Declaration

Figures adapted from Nakkiran et al. arXiv source. AI tools assisted with drafting and layout; the group retains responsibility for verifying all mathematical and source claims.